Lecture 6

Symmetry, conservation and degeneracy

Objectives. *The three-level system of Lecture 4 has a degenerate level, and Lecture 5 found two incompatible ways to split it. Why are levels degenerate at all? This lecture answers with symmetries. An observable that commutes with the Hamiltonian keeps its statistics for ever, and a unitary that commutes with it maps every motion to another motion. Two such unitaries that do not commute with each other force a degenerate level. A magnetic field breaks one symmetry of the three-level system and splits its level in two. Finally, the first-order shifts under a small perturbation are the eigenvalues of the perturbation inside the level, up to an error that the lecture bounds.*

By the end of this lecture you will be able to:

- Decide whether an observable is conserved, and whether a unitary is a symmetry, from its commutator with the Hamiltonian.
- Predict a degenerate level from two symmetries that do not commute.
- Find how a field or a strain splits the degenerate level of the three-level system, and which observables stay conserved.
- Find the first-order shifts of a level under a small perturbation, with the bound on their error.

6.1 Conserved quantities

Throughout this lecture the Hamiltonian H is constant, and $\rho_t = U_t \rho U_t^\dagger$ with $U_t = e^{-\mathrm{i}tH/\hbar}$. No measurement is made between the times 0 and t.

Definition 6.1 (Conserved observable)**.** An observable $A = \sum_a a\, E_a$ is *conserved* if, for every density matrix ρ, every reading a and every time t,

$$\mathrm{Tr}(\rho_t E_a) = \mathrm{Tr}(\rho E_a).$$

So the probability of each reading of a conserved observable never changes, whatever the state. By Theorem 4.4, the energy is conserved.

Theorem 6.2 (Conservation is commutation with H)**.** *An observable A is conserved if and only if $[A, H] = 0$.*

Proof. First, let $AH = HA$. By Lemma 4.3, applied to H, every E_k commutes with A. Applied again, to A and E_k, the lemma shows that every E_a commutes with every E_k, and so with

$U_t = \sum_k e^{-\mathrm{i}t\varepsilon_k/\hbar} E_k$. By the cyclicity of the trace,

$$\mathrm{Tr}(U_t \rho U_t^\dagger E_a) = \mathrm{Tr}(\rho U_t^\dagger E_a U_t) = \mathrm{Tr}(\rho E_a).$$

Conversely, let A be conserved. Then $\mathrm{Tr}\big(\rho(U_t^\dagger E_a U_t - E_a)\big) = 0$ for every ρ, and in particular $\big\langle \psi, (U_t^\dagger E_a U_t - E_a)\psi \big\rangle = 0$ for every unit vector ψ. The matrix in brackets is self-adjoint, so it is 0, as in the proof of Theorem 5.2. Hence $U_t^\dagger E_a U_t = E_a$ for every t. The derivative at $t = 0$ is $\frac{\mathrm{i}}{\hbar}(HE_a - E_a H) = 0$, so every E_a commutes with H, and so does $A = \sum_a a\, E_a$. ∎

By Theorem 5.2, a conserved observable is one that is compatible with the Hamiltonian. Its readings also survive the evolution. If a reading of A gives a at time 0, a second reading at time t gives a again with certainty, since the branch operator of the readings a and a' is $E_{a'} U_t E_a = U_t E_{a'} E_a = 0$ for $a' \neq a$.

By the definition, a conserved observable has a constant mean $\mathrm{Tr}(\rho_t A)$. The converse fails. A constant mean in one state, or even constant statistics in one state, does not make A conserved, since the definition asks for every state. Every observable has constant statistics in a stationary state.

6.2 Symmetries

Definition 6.3 (Symmetry)**.** A *symmetry* of the Hamiltonian H is a unitary matrix V with $VH = HV$.

By Lemma 4.3 and (4.3), a symmetry commutes with every U_t. Conversely, a unitary that commutes with every U_t commutes with H, as the derivative at $t = 0$ shows. So V is a symmetry if and only if $VU_t = U_t V$ for all t. In other words, if ψ_t is a *motion* of the system, that is, a solution of the Schrödinger equation, then so is $V\psi_t$. Preparing $V\psi$ and waiting for the time t gives $U_t V\psi = VU_t\psi$, the image under V of what ψ becomes. A symmetry also keeps the energy statistics, since $\mathrm{Tr}(V\rho V^\dagger E_k) = \mathrm{Tr}(\rho V^\dagger E_k V) = \mathrm{Tr}(\rho E_k)$.

Symmetries often come in families, such as rotations by every angle about one axis. The next proposition links such a family to a conserved observable.

Proposition 6.4 (A family of symmetries and its generator)**.** *Let A be an observable, and $V_s = e^{-\mathrm{i}sA/\hbar}$ for $s \in \mathbb{R}$.*

1. *All V_s are symmetries of H if and only if $[A, H] = 0$, that is, if and only if A is conserved.*
2. *A symmetry V with $V^2 = \mathbb{1}$ is self-adjoint, its readings are among $+1$ and -1, and it is conserved.*

Proof.

1. If $[A, H] = 0$, then by Lemma 4.3 H commutes with every projection E_a of A, and so with $V_s = \sum_a e^{-\mathrm{i}sa/\hbar} E_a$. Conversely, if $V_s H = HV_s$ for every s, the derivative at $s = 0$ gives $-\frac{\mathrm{i}}{\hbar}(AH - HA) = 0$. The rest is Theorem 6.2.
2. A unitary V with $V^2 = \mathbb{1}$ has $V^\dagger = V^{-1} = V$. Each eigenvalue λ of V satisfies $\lambda^2 = 1$. Since V commutes with H, it is conserved by Theorem 6.2. ∎

The observable A is called the *generator* of the family V_s. By Proposition 6.4, the generator of a family of symmetries is conserved. In classical mechanics the corresponding statement is Noether's theorem. We do not need it.

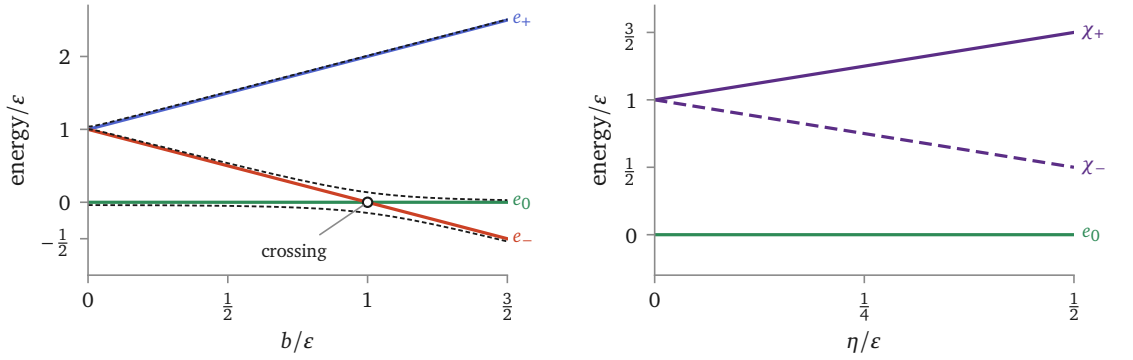


Figure 6.1: The levels of the three-level system of Lecture 4 with one symmetry broken (Example 6.6). Left: a field along the axis, $H + bZ$. The levels $\varepsilon \pm b$ and 0 belong to $e_\pm$ and e_0, and the lines of e_0 and e_- cross at $b = \varepsilon$. Dashed: a field slightly off the axis adds $cX_\perp$ with $c = 0.2\varepsilon$ (the box *The physics* below), and the two lines avoid each other (§ 6.6). Right: strain, $H + \eta X$. The levels $\varepsilon \pm \eta$ belong to $\chi_\pm$. Both panels show the ideal model of (4.1), without the smaller interactions of a real NV centre.

6.3 Symmetry forces degeneracy

Theorem 6.5 (Non-commuting symmetries force degeneracy)**.** *If a Hamiltonian H on $\mathbb{C}^n$ has two symmetries V and W with $VW \neq WV$, then some energy level of H is degenerate.*

Proof. Suppose every level of H is simple, with unit eigenvectors $u_1, \dots, u_n$. By the proof of Lemma 4.3, V maps each eigenspace of H into itself. Each eigenspace is the line through some u_k, so $Vu_k = v_k u_k$ for a number v_k. In the same way $Wu_k = w_k u_k$. Then $VWu_k = v_k w_k u_k = WVu_k$ for every k, so $VW = WV$, against the hypothesis. ∎

The theorem predicts a degeneracy from the symmetries alone, before the spectrum is computed. One symmetry, or several that commute with one another, forces nothing. For example, § 6.4 gives a Hamiltonian with a whole family of symmetries and three simple levels. Conversely, every Hamiltonian with a degenerate level has non-commuting symmetries. Inside a degenerate eigenspace, any two non-commuting unitaries of that eigenspace, extended by $\mathbb{1}$ on the other eigenspaces, are such symmetries. So the theorem is useful when the symmetries are known first, as they are when they come from the geometry of the system.

6.4 A three-level system in a field

Example 6.6 (The degenerate level, split)**.** Take the Hamiltonian $H = \varepsilon(|e_+\rangle\langle e_+| + |e_-\rangle\langle e_-|)$ of (4.1) and the observables Z and X of Lecture 4. Figure 6.1 shows the levels in a field and under strain.

1. *Two symmetries.* The matrices

$$V_\alpha = e^{-i\alpha Z} = \operatorname{diag}(e^{-i\alpha}, 1, e^{i\alpha}), \qquad \alpha \in \mathbb{R},$$

are symmetries of H, since V_α and H are both diagonal. The *mirror* M, with $Me_\pm = e_\mp$ and $Me_0 = e_0$, is a symmetry too. It exchanges e_+ and e_-, which have the same energy.

Since $MZM = -Z$, we get $MV_\alpha M = V_{-\alpha}$, and this differs from V_α unless $e^{2i\alpha} = 1$. So for $\alpha \notin \pi\mathbb{Z}$ the symmetries M and V_α do not commute, and by Theorem 6.5 H has a degenerate level. It is the level ε.

2. *A field along the axis.* A magnetic field along the axis adds bZ, where $b > 0$ is an energy proportional to the field:

$$H + bZ = \operatorname{diag}(\varepsilon + b,\ 0,\ \varepsilon - b).$$

Every V_α is still a symmetry, so, by Proposition 6.4, Z is conserved. The mirror is not, since $M(H + bZ)M = H - bZ$. The levels $\varepsilon + b$, 0 and $\varepsilon - b$ are distinct unless $b = \varepsilon$. A family of commuting symmetries remains, and it forces nothing. At $b = \varepsilon$ the states e_0 and e_- have the same energy 0. No symmetry of the whole family $H + bZ$ forces this, and such a degeneracy is called *accidental*.

3. *Strain instead.* A strain, a small deformation of the crystal, adds instead ηX, with $\eta > 0$. Since $MXM = X$, the mirror is still a symmetry. The rotations V_α with $\alpha \notin \pi\mathbb{Z}$ are not, since X does not commute with Z. The levels are $\varepsilon + \eta$ on χ_+, $\varepsilon - \eta$ on χ_-, and 0 on e_0. Now X is conserved, and Z is not. Since $M^2 = \mathbb{1}$, the mirror is itself a conserved observable, by Proposition 6.4. Its readings are +1 on χ_+ and e_0, and −1 on χ_-.

4. *Conserved and not conserved.* Under $H + bZ$, prepare χ_+. Then

$$U_t\chi_+ = \frac{1}{\sqrt{2}}\left(e^{-i(\varepsilon+b)t/\hbar}e_+ + e^{-i(\varepsilon-b)t/\hbar}e_-\right).$$

So a reading of Z gives ±1 with probability $\frac{1}{2}$ at every time. A reading of X gives +1 with probability

$$|\langle\chi_+, U_t\chi_+\rangle|^2 = \frac{1}{4}\left|e^{-ibt/\hbar} + e^{ibt/\hbar}\right|^2 = \cos^2(bt/\hbar).$$

The field turns χ_+ into χ_- and back, with the Bohr frequency $2b/\hbar$ of the split pair.

The example shows what the symmetries protect. The rotations about the axis keep Z conserved, and together with the mirror they force the degeneracy of e_+ and e_-. A field along the axis spares the rotations and breaks the mirror, and the level splits. Strain spares the mirror and breaks the rotations, and the level splits too. Lecture 7 identifies V_α as a rotation by the angle α about the axis, and M, up to a sign, as a rotation by π about a perpendicular axis.

The physics

In 1896 Pieter Zeeman saw the spectral lines of sodium broaden in a magnetic field. With finer instruments such lines are seen to split, and the splitting of spectral lines by a field carries his name. For the NV^- centre of Lecture 4, in the ideal model of Example 6.6 with the field B along the axis of the centre and strain neglected, $b = \hbar\gamma_e B$. Here γ_e is the size of the electron's gyromagnetic ratio, $\gamma_e/2\pi \approx 28.0$ GHz/T. The γ of Lecture 1 is negative for an electron, and $\gamma_e = |\gamma|$. For $0 \le b < \varepsilon$ the microwave resonance of Lecture 4 splits into two lines, at $D \pm (\gamma_e/2\pi)B$, so a field of 1 mT moves each line by 28 MHz. Measuring the two lines measures the field, and NV centres are used as small magnetometers. The model's crossing at $b = \varepsilon$ falls near $B = 102$ mT. There the real centre shows an anticrossing instead. Smaller interactions that the model leaves out, with the nitrogen nucleus among them, join the two levels and keep them apart, as § 6.6 explains. A field slightly off the axis does the same in the model.

It adds $cX_\perp$, with $c > 0$ and

$$X_\perp = \frac{1}{\sqrt{2}}(|e_+\rangle\langle e_0| + |e_0\rangle\langle e_+| + |e_0\rangle\langle e_-| + |e_-\rangle\langle e_0|),$$

which Lecture 7 identifies as $J_x/\hbar$. This term joins e_0 and e_-, with $\langle e_0, cX_\perp e_-\rangle = c/\sqrt{2}$, and for small c the two levels then avoid each other by about $\sqrt{2}\,c$ (Figure 6.1, dashed).

6.5 A small perturbation

The field and the strain of Example 6.6 could be solved exactly. Most perturbations cannot, but a small one can be handled with a bound on the error. Let H_0 and H_1 be self-adjoint, and let the Hamiltonian be

$$H = H_0 + H_1.$$

Let ε be a level of H_0 with spectral projection E of rank d. Let every other eigenvalue of H_0 be at a distance at least $g > 0$, the *gap*, from ε. For a self-adjoint A, let $\|A\|$ be the largest modulus of its eigenvalues, the operator norm of *Linear Algebra*, Def. 14.3. Expanding ϕ in an eigenbasis of A gives $\|A\phi\| \le \|A\|\,\|\phi\|$. As always, E is a projection, and an eigenvalue of H is written λ.

Lemma 6.7 (An eigenvalue near a number). *Let A be self-adjoint, μ real and $\phi \neq 0$. Then A has an eigenvalue λ with*

$$|\lambda - \mu| \le \frac{\|(A-\mu)\phi\|}{\|\phi\|}.$$

Proof. In an orthonormal eigenbasis, $\phi = \sum_j c_j v_j$ with $Av_j = \lambda_j v_j$, and

$$\|(A-\mu)\phi\|^2 = \sum_j |c_j|^2 (\lambda_j - \mu)^2 \ge \min_j (\lambda_j - \mu)^2 \,\|\phi\|^2 .$$

The eigenvalue λ_j that attains the minimum is the λ of the lemma. ∎

Theorem 6.8 (A small perturbation). *In this notation, let $\|H_1\| \le g/4$.*

1. *H has exactly d eigenvalues, counted with multiplicity, in $[\varepsilon - g/2, \varepsilon + g/2]$.*
2. *Each of them lies within $3\,\|H_1\|^2/g$ of $\varepsilon + v$ for some eigenvalue v of EH_1E on the range of E.*
3. *For each such v, some eigenvalue of H lies within $3\,\|H_1\|^2/g$ of $\varepsilon + v$.*

Proof. Let $\mathcal{W}$ be the span of the eigenvectors of H with eigenvalues in the interval. Its orthogonal complement is spanned by eigenvectors whose eigenvalues are farther than $g/2$ from ε.

1. If $\dim \mathcal{W} < d$, the range of E meets that complement in some $\phi \neq 0$, since their dimensions add up to more than n. Since $H_0\phi = \varepsilon\phi$, we have $\|(H-\varepsilon)\phi\| = \|H_1\phi\| \le \frac{g}{4}\|\phi\|$. This contradicts $\|(H-\varepsilon)\phi\| > \frac{g}{2}\|\phi\|$, which holds on the complement. If $\dim \mathcal{W} > d$, then $\mathcal{W}$ meets the range of $\mathbb{1} - E$ in some $w \neq 0$, for the same reason. On $\mathcal{W}$, $\|(H-\varepsilon)w\| \le \frac{g}{2}\|w\|$. On the range of $\mathbb{1} - E$, $\|(H_0-\varepsilon)w\| \ge g\,\|w\|$, so

   $$\|(H-\varepsilon)w\| \ge \|(H_0-\varepsilon)w\| - \|H_1 w\| \ge \frac{3g}{4}\,\|w\|\,,$$

   a contradiction.

2. Let $Hw = \lambda w$ with $w \neq 0$ and $|\lambda - \varepsilon| \leq g/2$. Since H_0 commutes with E, applying $\mathbb{1} - E$ to $(\lambda - H_0)w = H_1 w$ gives $(\lambda - H_0)(\mathbb{1} - E)w = (\mathbb{1} - E)H_1 w$. Every eigenvalue of H_0 on the range of $\mathbb{1} - E$ is at least $g/2$ from λ, so $\|(\mathbb{1} - E)w\| \leq \frac{2}{g}\|H_1\|\|w\| \leq \frac{1}{2}\|w\|$. Applying E instead gives $\big(EH_1E - (\lambda - \varepsilon)\big)Ew = -EH_1(\mathbb{1} - E)w$, whose norm is, by the previous bound, at most $\frac{2}{g}\|H_1\|^2\|w\|$. By Pythagoras, $\|Ew\|^2 = \|w\|^2 - \|(\mathbb{1} - E)w\|^2 \geq \frac{3}{4}\|w\|^2$, so $\|Ew\| \geq \frac{\sqrt{3}}{2}\|w\|$. Lemma 6.7, for EH_1E on the range of E, with $\mu = \lambda - \varepsilon$ and the vector Ew, gives an eigenvalue v of EH_1E with

$$|\lambda - \varepsilon - v| \leq \frac{4}{\sqrt{3}}\frac{\|H_1\|^2}{g},$$

and $\frac{4}{\sqrt{3}} < 3$.

3. Let u be a unit vector in the range of E with $EH_1Eu = vu$, so that $|v| \leq \|H_1\|$ and $H_1u = vu + (\mathbb{1} - E)H_1u$. Let $R = \sum_k E_k/(\varepsilon - \varepsilon_k)$, summed over the other levels of H_0. Then $(H_0 - \varepsilon)R = -(\mathbb{1} - E)$ and $\|R\chi\| \leq \|\chi\|/g$ for every χ. Consider the vector $\phi = u + R(\mathbb{1} - E)H_1u$. Its second term is orthogonal to u, so $\|\phi\| \geq 1$. Since $(H_0 - \varepsilon)u = 0$,

$$(H - \varepsilon - v)\phi = (H_1 - v)R(\mathbb{1} - E)H_1u.$$

Since $\|H_1 - v\| \leq 2\|H_1\|$, the norm of this vector is at most $2\|H_1\|^2/g$. Lemma 6.7 gives the eigenvalue. ∎

This is *first-order perturbation theory*. For a simple level with unit eigenvector u, the level moves to $\varepsilon + \langle u, H_1u\rangle$, up to an error of at most $3\|H_1\|^2/g$.

Remark 6.9 (Second order). For a simple level, take an eigenvector w of H for the eigenvalue λ within $g/2$ of ε, as in the proof of Theorem 6.8, and scale it so that $Ew = u$. On the range of $\mathbb{1} - E$, $\|(\lambda - (\mathbb{1} - E)H(\mathbb{1} - E))\chi\| \geq \frac{g}{4}\|\chi\|$, so the matrix $\lambda - (\mathbb{1} - E)H(\mathbb{1} - E)$ has an inverse S there. The $\mathbb{1} - E$ part of $Hw = \lambda w$ gives $(\mathbb{1} - E)w = S(\mathbb{1} - E)H_1u$, and the inner product of the equation $Hw = \lambda w$ with u gives exactly

$$\lambda = \varepsilon + \langle u, H_1u\rangle + \langle (\mathbb{1} - E)H_1u, S(\mathbb{1} - E)H_1u\rangle\,.$$

Replacing λ by ε and H by H_0 inside S gives the *second-order formula*

$$\varepsilon + \langle u, H_1u\rangle + \sum_j \frac{\big|\langle u_j, H_1u\rangle\big|^2}{\varepsilon - \varepsilon_j},$$

with u_j an orthonormal eigenbasis of H_0 for the other levels, $H_0u_j = \varepsilon_j u_j$. Exercise Sheet 6 shows that it is within $4\|H_1\|^3/g^2$ of λ.

Example 6.10 (Field and strain, to first order). Take $H_0 = H$ of (4.1) and its level ε, with $E = |e_+\rangle\langle e_+| + |e_-\rangle\langle e_-|$. So $d = 2$ and $g = \varepsilon$. Let $H_1 = bZ + \eta X$. In the basis e_+, e_- of the range of E, the matrix of EH_1E is

$$\begin{pmatrix} b & \eta \\ \eta & -b \end{pmatrix},$$

with the eigenvalues $\pm\sqrt{b^2 + \eta^2}$. To first order, the level splits into $\varepsilon \pm \sqrt{b^2 + \eta^2}$. Here this is exact, since H_1 does not join e_0 to the pair (Exercise Sheet 6). A perturbation with $EH_1E = 0$ does not move this level at first order. It may still split it at second order (Exercise Sheet 6).

6.6 Two levels that do not cross

Proposition 6.11 (Two levels). *The self-adjoint matrix* $\begin{pmatrix} a_1 & v \\ \bar{v} & a_2 \end{pmatrix}$, *with* a_1, a_2 *real, has the eigenvalues*

$$\frac{a_1 + a_2}{2} \pm \sqrt{\frac{(a_1 - a_2)^2}{4} + |v|^2},$$

which differ by at least $2\,|v|$.

Proof. The eigenvalues are the roots of the characteristic polynomial $\lambda^2 - (a_1+a_2)\lambda + a_1a_2 - |v|^2$, and $(a_1 + a_2)^2/4 - a_1a_2 + |v|^2 = (a_1 - a_2)^2/4 + |v|^2$. ∎

Without the coupling v the levels are a_1 and a_2, and they cross when a parameter moves a_1 through a_2. With $v \neq 0$ they never come closer than $2\,|v|$. The levels *avoid* each other. Far from the crossing, for $|v|$ much smaller than $a_1 - a_2 > 0$, the upper level is $a_1 + |v|^2/(a_1 - a_2)$ to second order, as in Remark 6.9. At $a_1 = a_2$ the levels are $a_1 \pm |v|$.

In Example 6.6 the levels of e_0 and e_- do cross, at $b = \varepsilon$. They may cross. The observable Z is conserved, the two states have different readings of Z, and so nothing in $H + bZ$ joins them. Levels with different readings of a conserved observable may cross. Proposition 6.11 shows how a coupling keeps two levels apart, an instance of the non-crossing rule of von Neumann and Wigner, named here and not proved.

6.7 Symmetries of a ring

Lecture 8 studies a particle that hops between $N \geq 3$ sites on a ring. Its states are the rays of $\mathbb{C}^N$, with an orthonormal basis e_x, one for each site, labelled by $x \in \mathbb{Z}_N$, so that $e_{x+N} = e_x$. Two unitaries describe the geometry of the ring. The *shift* T moves the particle one site forward, and the mirror M reflects the ring:

$$Te_x = e_{x+1}, \qquad Me_x = e_{-x}.$$

The Hamiltonian of Lecture 8 is

$$H = c\mathbb{1} - \kappa(T + T^\dagger),$$

with real numbers c and $\kappa > 0$. Both T and M are symmetries of H. The shift commutes with $T + T^\dagger$, since $T^\dagger = T^{-1}$. For the mirror,

$$MTMe_x = MTe_{-x} = Me_{1-x} = e_{x-1},$$

so $MTM = T^{-1} = T^\dagger$, and therefore $M(T + T^\dagger)M = T^\dagger + T$.

The shift and the mirror do not commute, since $MTe_0 = Me_1 = e_{-1}$, while $TMe_0 = Te_0 = e_1$, and $e_{-1} \neq e_1$ for $N \geq 3$. By Theorem 6.5, the ring has a degenerate level, and we know this before computing a single energy. Lecture 8 finds all the levels and which of them are degenerate.

Symmetries of an n-level system

Conserved: an observable A keeps its statistics for every state and time if and only if $[A, H] = 0$ (Theorem 6.2). *Symmetry:* a unitary V with $VH = HV$. It maps motions to motions. The generator of a family of symmetries is conserved (Proposition 6.4). *Degeneracy:* two symmetries that do not commute force a degenerate level (Theo-

rem 6.5). A perturbation that breaks one of them may split it. *Perturbation:* an H_1 with $\|H_1\| \le g/4$ shifts a level ε by the eigenvalues of EH_1E, up to $3\|H_1\|^2/g$ (Theorem 6.8).

Summary.

- *Conservation is commutation with H*: an observable keeps its statistics in every state exactly when it commutes with the Hamiltonian (Theorem 6.2).
- *A family of symmetries and its generator*: the generator of a family of symmetries is conserved (Proposition 6.4).
- *Non-commuting symmetries force degeneracy* (Theorem 6.5).
- *A small perturbation*: for $\|H_1\| \le g/4$, a level of rank d becomes d levels near ε plus the eigenvalues of EH_1E, up to $3\|H_1\|^2/g$ (Theorem 6.8).
- *Two levels*: a coupling v keeps two levels at least $2|v|$ apart (Proposition 6.11).

Lecture 7 finds the multiplets that the rotations, which do not commute, force on a system.

References

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